

Mathematics (Science)	Tenth, 2024 Lahore Board	Group - I
Time: 2.10 hrs.	Essay Type	Marks : 60

PART - I

Q.2 Write short answers to any SIX (6) questions: 12

- What is meant by radical equation?
- Write the quadratic equation in standard form:
$$\frac{x}{x+1} + \frac{x+1}{x} = 6$$
- Solve by factorization : $x^2 - x - 20 = 0$
- Find the discriminant of the given quadratic equation:
 $16x^2 - 24x + 9 = 0$
- Evaluate : $\omega^{-13} + \omega^{-17}$
- Write the quadratic equation having following roots :
1, 5
- If $R \propto T^2$ and $R = 8$ when $T = 3$, find k
- Define ratio and give one example.
- Find a third proportion to : 6, 12

Q.3 Write short answers to any SIX (6) questions: 12

- What is meant by an Identity?
- Convert into proper fraction : $\frac{6x^4}{x^3 + 1}$
- If A and B are two sets, then represent $A - B$ in set builder notation.
- If $Y = Z^+$, $T = O^+$, then find $Y \cup T$
- Find number of elements in $Y \times X$, if $X = \{a, b, c\}$ and $Y = \{d, e\}$
- Find $A \times B$ if $A = \{a, b\}$, $B = \{c, d\}$
- Define median.
- Using logarithmic formula, find the geometric mean of 2, 4, 8
- Find range : 11500, 12400, 15000, 14500, 14800

Q.4 Write short answers to any SIX (6) questions: 12

- Define coterminal angles.
- Convert -225° to radian.
- Find r , when $\theta = 75^\circ$, $\ell = 52\text{cm}$.
- Show that : $\sec\theta - \cos\theta = \tan\theta \sin\theta$
- Define exterior of a circle.
- Define circumference.
- Define cyclic quadrilateral.
- Draw a circle of radius 4 cm passing through points A and B, 5 cm apart.
- Define escribed circle.

PART-II

Note: Attempt any THREE questions in all. But question No. 9 is Compulsory.

Q.5(a) Solve the equation $x^2 - 3x - 4 = 0$ by completing square.
(b) Prove that : $x^3 + y^3 + z^3 - 3xyz = (x + y + z)(x + wy + w^2z)(x + w^2y + wz)$

Q.6(a) Using theorem of componendo-dividendo, find the value of

$$\frac{m+5n}{m-5n} + \frac{m+5p}{m-5p} \text{ if } m = \frac{10np}{n+p}$$

(b) Resolve into partial fractions : $\frac{3x+7}{(x+3)(x^2+4)}$

Q.7(a) If $U = \{1, 2, 3, 4, \dots, 10\}$, $A = \{2, 4, 6, 8, 10\}$, $B = \{2, 3, 5, 7\}$ then verify $(A \cup B)' = A' \cap B'$

(b) Find mean

Classes	33-40	41-50	51-60	61-70	71-75
No. of students	28	31	12	9	5

Q.8(a) If $\cos\theta = \frac{-2}{3}$ and terminal arm of the angle θ is in quadrant III. Find the values of remaining trigonometric functions.

(b) Inscribe a circle in a triangle ABC with sides :
 $|AB| = 6\text{cm}$, $|BC| = 3\text{cm}$, $|CA| = 4\text{cm}$

Q.9 Prove that a straight line drawn from the centre of a circle to bisect a chord (which is not a diameter) is perpendicular to the chord.

OR.

Prove that the measure of a central angle of a minor arc of a circle, is double that of angle subtended by corresponding major arc.